SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	K. Srinivasa Reddy	**Reg. No.:**	192525452
Section / Batch:	B. Tech AIML	**Date:**	08/07/2026

Code of Conduct

I K. Srinivasa Reddy(19252542)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: k. srinivasa reddy

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
		GRAND TOTAL:	**100 Marks**

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QUESTIONS

Q1
C Code – Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q6
C Code – Linked List Traversal
10 marks

Q1 C Code – Pointer with Array

10 marks

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

Your Answer

30

Save Answer

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C Code - Pointer Increment
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Q5
C Code - Array Address
10 marks

Q6
C Code – Linked List Traversal
10 marks

Q7
C Code - Recursion (Factorial)
10 marks

Q8
C Code - Recursion (Sum)
10 marks

Q2 C Code - Linked List Node

10 marks

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

20

Save Answer

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- **Q4** C Code · Pointer Increment · 10 marks
- **Q5** C Code · Array Address · 10 marks

Q3 **C Code - Binary Search Complexity** 10 marks

```
while(low<=high){
mid=(low+high)/2;
}
```

Your Answer

```
while (low <= high) {
    mid = (low + high) / 2;
}
```

Save Answer

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- **Q5** C Code · Array Address · 10 marks
- **Q6** C Code · Linked List Traversal · 10 marks
- **Q7**

Q4 **C Code - Pointer Increment** 10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

```
10
```

Save Answer

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- **Q4** C Code · Pointer Increment · 10 marks
- **Q5** C Code · Array Address · 10 marks

Q5 **C Code - Array Address** 10 marks

```
int arr[]={5,10,15};
printf("%d",*(arr+1));
```

Your Answer

```
10
```

Save Answer

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10 marks

Q5
C Code – Array Address
10 marks

Q6

Q6 C Code – Linked List Traversal

10 marks

```
Node1 -> Node2 -> Node3 (Node values are 10,20,30)
current=head;
current=current->next;
printf("%d",current->data);
```

Your Answer

```
20
```

Save Answer

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C Code – Array Address
10 marks

Q6
C Code – Linked List Traversal
10 marks

Q7
C Code – Recursion (Factorial)
10 marks

Q7 C Code – Recursion (Factorial)

10 marks

```
#include <stdio.h>
int fact(int n){
if(n==0)
return 1;
return n * fact(n-1);
}
int main(){
printf("%d", fact(4));
return 0;
}
```

Your Answer

```
#include <stdio.h>
int fact(int n){
if(n==0)
return 1;
return n * fact(n-1);
}
```

Save Answer

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C Code - Array Address
10 marks

Q6
C Code - Linked List Traversal
10 marks

Q7
C Code - Recursion (Factorial)
10 marks

Q8 **C Code – Recursion (Sum)** 10 marks

```
#include <stdio.h>
int sum(int n){
if(n==0)
return 0;
return n + sum(n-1);
}
int main(){
printf("%d", sum(5));
}
```

Your Answer

15

Save Answer

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10 marks

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C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q6
C Code - Linked List Traversal
10 marks

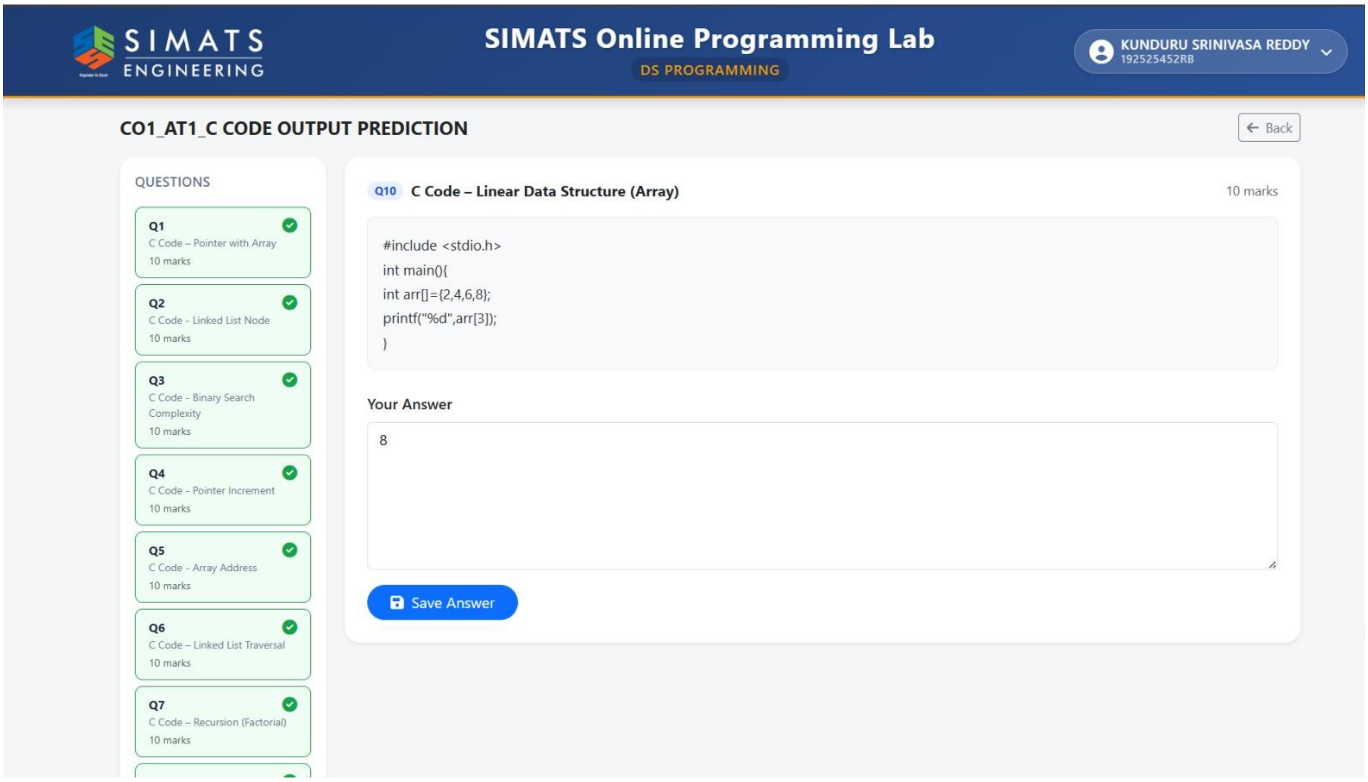
Q9 **C Code – Primitive Data Type** 10 marks

```
#include <stdio.h>
int main(){
int a=10;
char ch='A';
printf("%d %c",a,ch);
}
```

Your Answer

10A

Save Answer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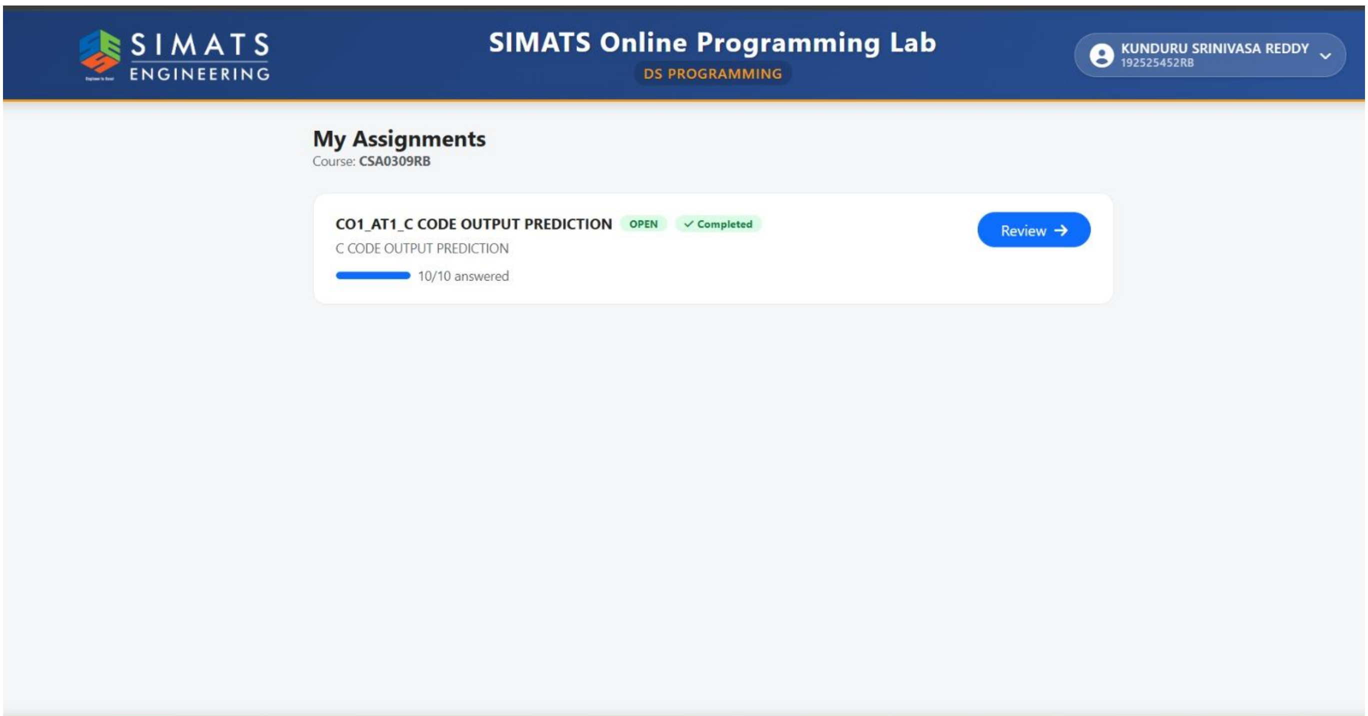
10 marks
Q5
C Code - Array Address
10 marks
Q6
C Code – Linked List Traversal
10 marks
Q7
C Code – Recursion (Factorial)
10 marks
Q10
C Code – Linear Data Structure (Array)
10 marks
#include <stdio.h>
int main(){
int arr[]={2,4,6,8};
printf("%d",arr[3]);
}
Your Answer
8
Save Answer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Course: CSA0309RB
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OPEN
✓ Completed
C CODE OUTPUT PREDICTION
10/10 answered
Review